

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data to Delta Tables

Title: Load Customer and Order Data to Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- Delta tables for customer and order data exist or can be created

Main Flow / Functional Steps:

- Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and load it into a Delta table named "customer"
- Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` and load it into a Delta table named "order"
- Add a new column "TotalAmount" to the "order" table, calculated as the product of "PricePerUnit" and "Qty"

FR-DLT-002: Clean and Process Customer and Order Data

Title: Clean and Process Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order Delta tables.

Preconditions:

- Customer and order Delta tables exist and contain data

Main Flow / Functional Steps:

- Remove null records from both "customer" and "order" tables
- Remove duplicate records from both "customer" and "order" tables

FR-DLT-003: Create Ordersummary Table and Load Data

Title: Create Ordersummary Table and Load Data

Description: The system must create an "ordersummary" table if it does not exist and load data into it by joining customer and order data.

Preconditions:

- Customer and order Delta tables exist and contain clean data
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

- Create "ordersummary" table in catalog "gen_ai_poc_databrickscoe" and schema "sdhc_wizard" if it does not exist
- Join "customer" and "order" tables on "CustId" and load the resulting data into the "ordersummary" table as an SCD type 2 table

FR-DLT-004: Update Ordersummary Table on Customer Changes

Title: Update Ordersummary Table on Customer Changes

Description: The system must update the "ordersummary" table whenever there are changes to the customer data.

Preconditions:

- "ordersummary" table exists and contains data
- Customer data has changed

Main Flow / Functional Steps:

- Identify changes to customer data
- Update "ordersummary" table by making old records inactive and new records active
- Update StartDate and EndDate accordingly to maintain history

FR-DLT-005: Create Customer Aggregate Spend Table and Load Data

Title: Create Customer Aggregate Spend Table and Load Data

Description: The system must create a "customeraggregatespend" table and load aggregated data into it.

Preconditions:

- "ordersummary" table exists and contains data
- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

- Create "customeraggregatespend" table in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist
- Aggregate "TotalAmount" from "ordersummary" table by "Name" and "Date"
- Load the aggregated data into the "customeraggregatespend" table

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement the above functionality using Delta Live Tables.

Preconditions:

- All previous steps have been completed successfully

Main Flow / Functional Steps:

- Implement steps 1 to 5 using Delta Live Tables to ensure data is processed and loaded correctly.